

Generally allows us to relate the magnitude of $\vec{h}$ to $\vec{r}$ and $\vec{v}$

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin \theta_{ab}$$

$$h = \|\vec{r} \times \vec{v}\| = rv \sin \phi \\ = rv \cos \gamma$$

Specific angular momentum is conserved...

$$\frac{d\vec{h}}{dt} = 0 \\ = \frac{d}{dt}(\vec{r} \times \vec{v}) = \frac{d}{dt}(\vec{r} \times \dot{\vec{r}}) = \cancel{\dot{\vec{r}} \times \dot{\vec{r}}} + \vec{r} \times \ddot{\vec{r}} = \vec{r} \times \left(-\frac{\mu}{r^3}\right) \vec{r} \quad \text{also } \vec{v}$$

Thus... $\frac{d\vec{h}}{dt} = 0$ so angular momentum is conserved.

Trajectory equation

$$r = \frac{p}{1 + e \cos \theta}$$

Our objective is to solve the two-body EDM.

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r} \quad \text{or} \quad \ddot{\vec{r}} = -\frac{\mu}{r^2} \hat{r}, \quad \hat{r} = \frac{\vec{r}}{r}, \quad r = \|\vec{r}\|$$

Using our knowledge that $\vec{h}$ is constant,

Therefore, take the cross product of the EDM with $\vec{h}$

$$\ddot{\vec{r}} \times \vec{h} = -\frac{\mu}{r^3} \vec{r} \times \vec{h} = -\frac{\mu}{r^3} \vec{r} \times (\vec{r} \times \dot{\vec{r}})$$

Recall: $\vec{h}$ is constant and

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

$\vec{h}$ is
constant $\downarrow$

$$\frac{d}{dt} (\dot{\vec{r}} \times \vec{h}) = -\frac{\mu}{r^3} \left[\underbrace{(\vec{r} \cdot \dot{\vec{r}})}_{r\dot{r}} \vec{r} - \underbrace{(\vec{r} \cdot \vec{r})}_{r^2} \dot{\vec{r}} \right]$$

$$r = \|\vec{r}\|$$

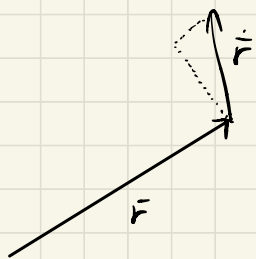
$$\dot{r} = \frac{d}{dt} \|\vec{r}\| = \frac{d}{dt} [\vec{r} \cdot \vec{r}]^{1/2} = \frac{1}{2} (\vec{r} \cdot \vec{r})^{-1/2} \frac{d}{dt} (\vec{r} \cdot \vec{r})$$

$$\dot{r} \neq v$$

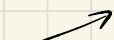
$$= \frac{1}{2} (\vec{r} \cdot \vec{r})^{1/2} (\dot{\vec{r}} \cdot \vec{r} + \vec{r} \cdot \dot{\vec{r}})$$

$$= \frac{1}{2} \left(\frac{1}{r} \right) 2\vec{r} \cdot \dot{\vec{r}} = \frac{1}{r} (\vec{r} \cdot \dot{\vec{r}})$$

$$\dot{r} = \frac{1}{r} (\vec{r} \cdot \dot{\vec{r}}) \Rightarrow \underline{\underline{r\dot{r} = \vec{r} \cdot \dot{\vec{r}}}}$$



$$\begin{aligned} \dot{r} &= \hat{r} \cdot \dot{\vec{r}} \\ \dot{r} &= \frac{\vec{r}}{r} \cdot \dot{\vec{r}} \end{aligned}$$



$$\frac{d}{dt} (\dot{\vec{r}} \times \vec{h}) = -\frac{\mu}{r^3} (r\dot{r}\vec{r} - r^2\dot{\vec{r}}) = -\mu \left(\frac{\dot{r}}{r^2} \vec{r} - \frac{1}{r} \dot{\vec{r}} \right)$$

Let's try to make some geometric sense of this...

$$\frac{d}{dt} (\hat{r}) = \frac{d}{dt} \left(\frac{\vec{r}}{r} \right) = \frac{d}{dt} (\vec{r} \cdot \vec{r})^{-1/2} \vec{r}$$

$$= \underbrace{-\frac{1}{2} (\vec{r} \cdot \vec{r})^{-3/2}}_{\frac{1}{r^3}} \underbrace{(\dot{\vec{r}} \cdot \vec{r} + \vec{r} \cdot \dot{\vec{r}})}_{(2\vec{r} \cdot \dot{\vec{r}})} \vec{r} + \underbrace{(\vec{r} \cdot \vec{r})^{-1/2}}_{\frac{1}{r}} \dot{\vec{r}}$$

$$= -\frac{\dot{r}}{r^2} \vec{r} + \frac{1}{r} \dot{\vec{r}} \quad \underline{\underline{r\dot{r}}}$$

Substituting...

$$\frac{d}{dt}(\dot{\vec{r}} \times \vec{h}) = \mu \frac{d}{dt} \left(\frac{\vec{r}}{r} \right)$$

Integrate both sides...

$$\dot{\vec{r}} \times \vec{h} = \mu \frac{\vec{r}}{r} + \vec{B} \quad \leftarrow \vec{B} \text{ is a constant of integration}$$

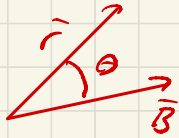
Take the dot product from the left with $\vec{r}$

$$\vec{r} \cdot (\dot{\vec{r}} \times \vec{h}) = \frac{\mu}{r} \underbrace{\vec{r} \cdot \vec{r}}_{r^2} + \vec{r} \cdot \vec{B} \quad \text{recall } \vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C}$$

$$\underbrace{(\vec{r} \times \dot{\vec{r}})}_{\vec{h}} \cdot \vec{h} = \frac{\mu}{r} r^2 + \vec{r} \cdot \vec{B}$$

h^2

$\vec{r} \cdot \vec{B}$



$r B \cos \theta$

$$h^2 = \mu r + r B \cos \theta$$

$$h^2 = r \left(\mu + B \cos \theta \right)$$

$$\frac{h^2}{\mu} = r \left(1 + \frac{B}{\mu} \cos \theta \right)$$

$$r = \frac{\frac{h^2}{\mu}}{1 + \frac{B}{\mu} \cos \theta}$$

← Two body Trajectory equation in polar coordinates.

Recall from geometry that the general equation of a conic section in polar coordinates...

$$r = \frac{p}{1 + e \cos \theta} \quad \text{where } \theta \text{ is defined as the angle between } \vec{r} \text{ and the point on the ellipse closest to the focus at the origin.}$$

p is the semi-latus rectum (semi-parameter)
 e is the eccentricity

The trajectory equation from before is clearly the same form...

$$p = \frac{h^2}{\mu} \quad e = \frac{B}{\mu} \Rightarrow \text{define a constant vector}$$

$$\vec{e} = \frac{1}{\mu} \vec{B} \quad (\text{eccentricity vector})$$

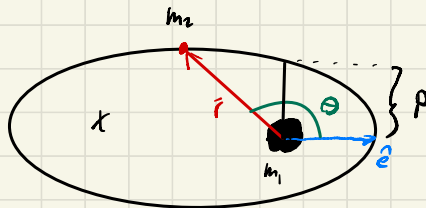
$$\vec{e} = \frac{1}{\mu} (\vec{r} \times \vec{h} - \mu_f \vec{r}) \rightarrow \text{constant}$$

Thus...

$$r = \frac{p}{1 + e \cos \theta} \quad \text{orbits for two-body problem are conic sections}$$

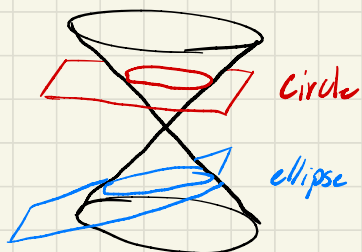
There are four types of (non-degenerate) conic sections

$e = 0$ circle
 $0 < e < 1$ ellipse
 $e = 1$ parabola
 $e > 1$ hyperbola



Conic Sections

A conic section or conic is the curve formed by the intersection of a plane with a cone. We assume a right-circular cone...



You can also write a conic as the solution to the quadratic equation in two variables...

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

$$\frac{(x-x_c)^2}{a^2} + \frac{(y-y_c)^2}{b^2} = 0$$

Note: Planes that pass through the vertex of the cone form the degenerate conics, which do occasionally occur.

In polar coordinates, we have $r = \frac{p}{1 + e \cos \theta}$

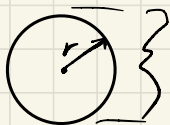
↳ which is related to cartesian coordinates $\bar{r} = \sqrt{x^2 + y^2}$

$$r^2 = x^2 + y^2 \quad x = r \cos \theta \quad y = r \sin \theta \quad (\text{in polar form})$$

There are a few key geometric relations you should know...

Circle

$$e = 0$$



$$d = 2r = 2a = 2p$$

$$e = 0$$

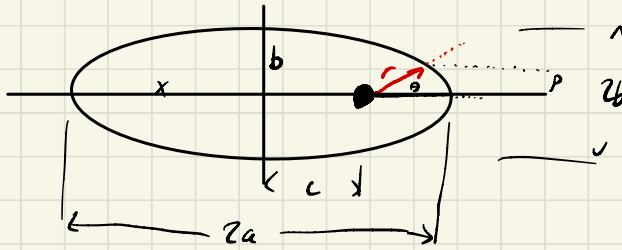
$$r = a = p$$

a = Semi-major axis

b = Semi-minor axis

ellipse

$$0 < e < 1$$



$$e = \frac{c}{a}$$